AI-Powered RAG Chatbot

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Introduction

- Internship focused on building an enterprise-grade AI-Powered Retrieval-Augmented Generation (RAG) Chatbot system.

- Purpose: Enable organizations to query their own documents using natural language and receive AI-powered answers with source citations.

- Domain: Artificial Intelligence / Natural Language Processing, specifically RAG-based conversational AI for enterprise data.

- Combines modern full-stack web development with cutting-edge AI/ML techniques including agentic reasoning, vector search, and multimodal document understanding.

Organization profile

- Company Name: Scanntek IT Solutions LLP
- Location: Kinfra IT Park, Calicut, Kerala
- Industry Domain: Digital Marketing, Web Development & IT Services
- Company Type: Product & Service Based (LLP, Inc. Nov 2022)
- Description: Scanntek IT Solutions LLP is an active, Kerala-based Limited Liability Partnership focusing on digital marketing, branding, web development, and IT services.
- Role: Full-Stack AI Developer — Designed and developed the end-to-end RAG Chatbot platform including backend API, frontend UI, AI pipeline, and DevOps infrastructure.

LEARNING MODULES

Concepts Learnt:

- Retrieval-Augmented Generation (RAG) Architecture
- Vector Embeddings & Semantic Search (3072-dim vectors)
- Agentic AI — ReAct Loop (Think → Action → Observe)
- LLM Prompt Engineering & Chain-of-Thought Reasoning
- Multimodal AI — Vision & Document Understanding
- Full-Stack Development with Django & React

Tools Utilized:

- LangChain, Qdrant, Google Gemini, Docker, PostgreSQL, Redis
- Git/GitHub, VS Code, Postman, Swagger UI

Period of Study: 10 Weeks (February – April 2026)

PROJECT(S) IMPLEMENTED

Title: AI-Powered RAG Chatbot

An enterprise-grade Retrieval-Augmented Generation chatbot for company database analysis, built with Django REST Framework and React.

Key Capabilities:

- RAG Chat with SSE-streamed responses & source citations
- Multimodal support (images, PDFs, DOCX, CSV, Markdown)
- Agentic semantic search with ReAct reasoning loop
- Document collections, tags, and scoped querying
- Admin dashboard with usage analytics & metrics
- JWT authentication, rate limiting, CSP/HSTS security
- Dockerized microservices with CI/CD pipelines

Problem Statement(s)

What problem is being solved?

- Organizations struggle to extract insights from large volumes of unstructured documents (PDFs, reports, spreadsheets).
- Traditional keyword search fails to understand context and semantic meaning, returning irrelevant results.
- Employees waste hours manually searching through documents.

Why is it important?

- Enterprises need instant, accurate, AI-powered answers grounded in their own data — not generic LLM hallucinations.
- RAG ensures responses are factual, cited, and traceable to source documents, enabling trust and compliance.

Objectives

General Objectives:

- Build an AI chatbot that retrieves and reasons over documents
- Provide accurate, citation-backed answers via RAG architecture

Specific Objectives:

- Implement multi-format document ingestion (PDF, DOCX, CSV, images)
- Build 3072-dim vector search with Qdrant + Gemini Embeddings
- Develop an agentic ReAct loop for intelligent query processing
- Create a real-time SSE chat interface with React 19
- Build an admin dashboard for usage analytics & user management
- Implement enterprise security: JWT, rate limiting, CSP headers
- Containerize the full stack with Docker Compose

Scope of the Project

What the project covers:

- Full-stack web application (Django + React)
- AI/ML pipeline: ingestion → embedding → retrieval → generation
- Multimodal RAG: text documents + image understanding
- Role-based access control (Admin, Editor, Viewer)
- CI/CD pipelines with GitHub Actions
- Dockerized microservices architecture

Limitations:

- Depends on Google Gemini API (cloud-based, requires API key)
- Not designed for real-time collaborative document editing
- Maximum context window constrained by LLM token limits

Technologies Used

Languages:

- Python 3.11, TypeScript 5, SQL

Frameworks:

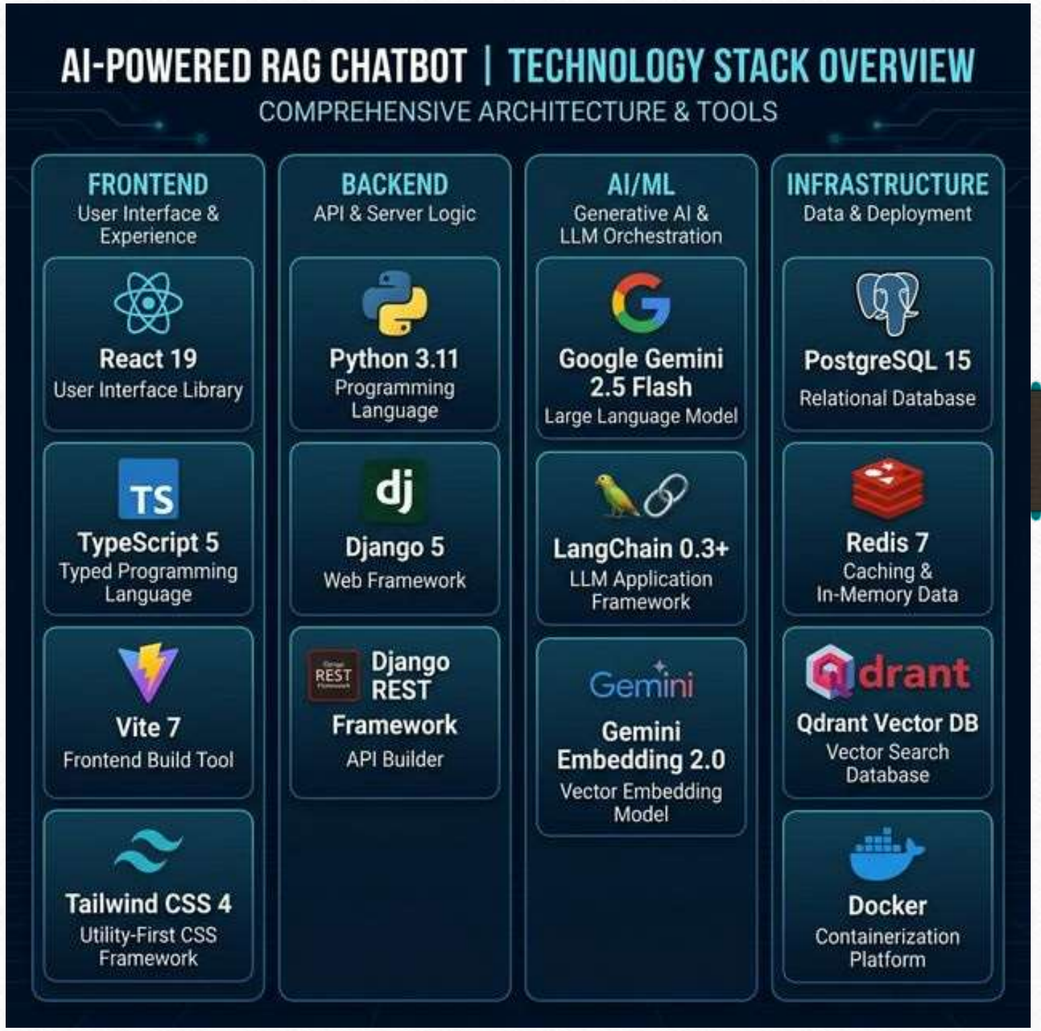
- Django 5, DRF, React 19, Vite 7
- Tailwind CSS 4

AI/ML:

- Google Gemini 2.5 Flash
- Gemini Embedding 2.0
- LangChain 0.3+

Infrastructure:

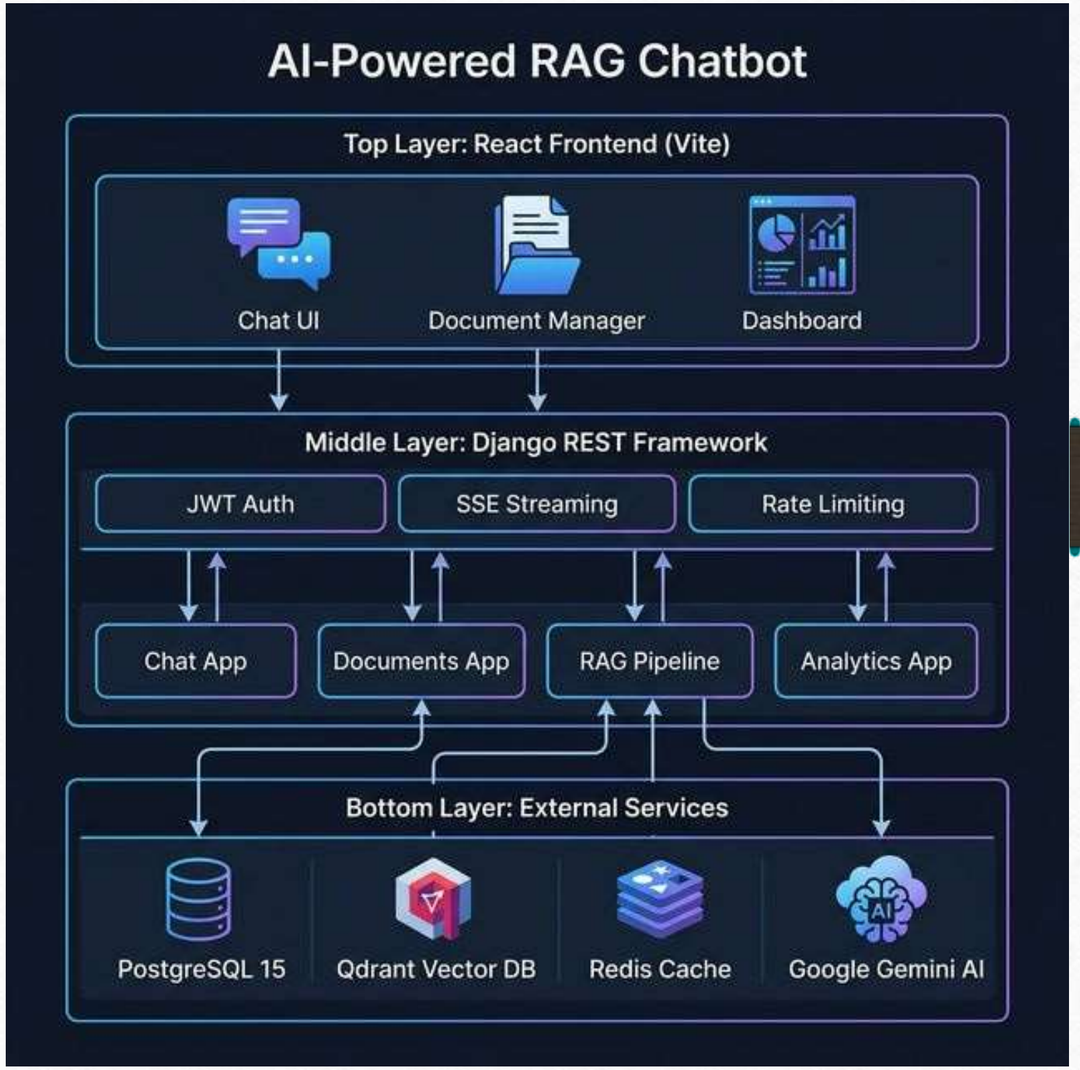
- PostgreSQL 15, Qdrant, Redis 7
- Docker, GitHub Actions CI/CD



System Architecture

Layered microservices architecture:

- Frontend → Backend API → App Modules → Storage Layer
- RAG Pipeline: Ingest → Embed → Store → Retrieve → Generate



Methodology

Development Process:

Agile (5 Sprints × 2 Weeks)

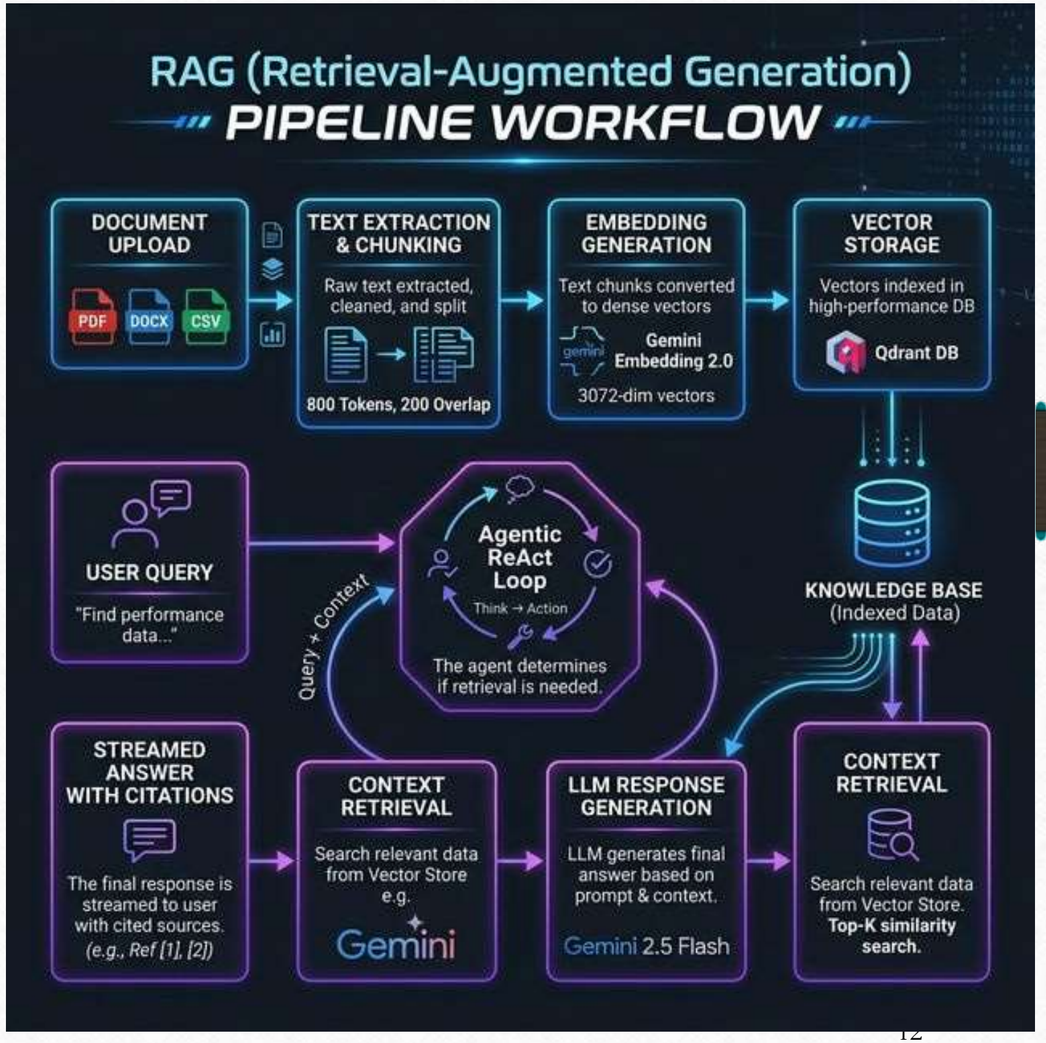
Sprint 1 — Requirements, setup, DB schema design

Sprint 2 — RAG pipeline (ingest, embed, vector storage)

Sprint 3 — Agentic reasoning, multimodal support

Sprint 4 — Dashboard, analytics, auth & security

Sprint 5 — Docker, CI/CD, testing, optimization



Implementation

Key Modules Implemented:

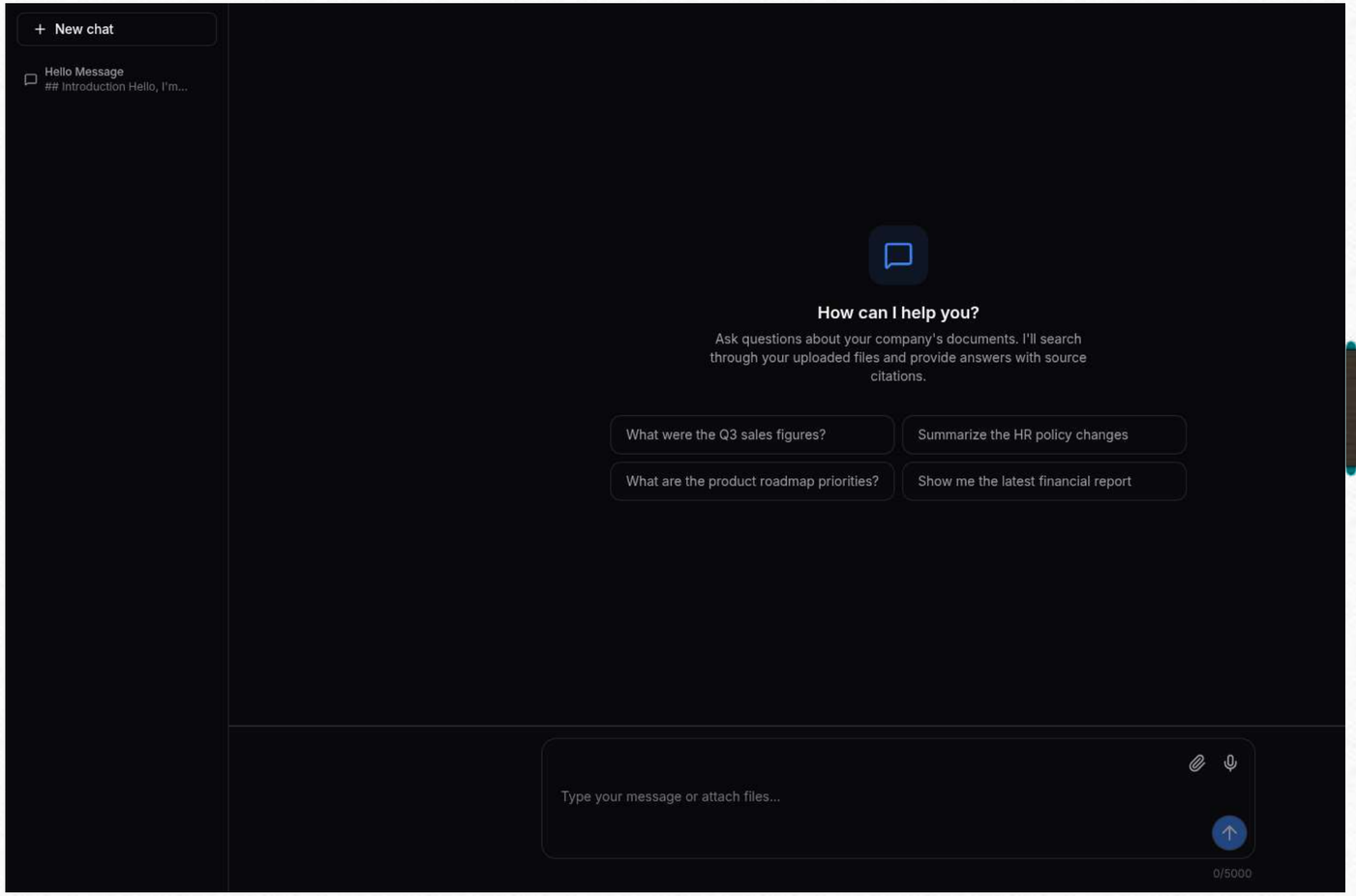
1. Document Management
   Upload PDF, DOCX, CSV, images with auto-chunking

2. RAG Pipeline
   LangChain + Qdrant + Gemini

3. Agentic Search
   ReAct: Think → Action → Observe

4. Multimodal AI
   Gemini Vision for images/charts

5. Real-time Chat (SSE streaming)
6. Admin Dashboard & Analytics
7. JWT Auth & Security Layer
8. Docker & CI/CD Pipeline



PROGRESS STATUS

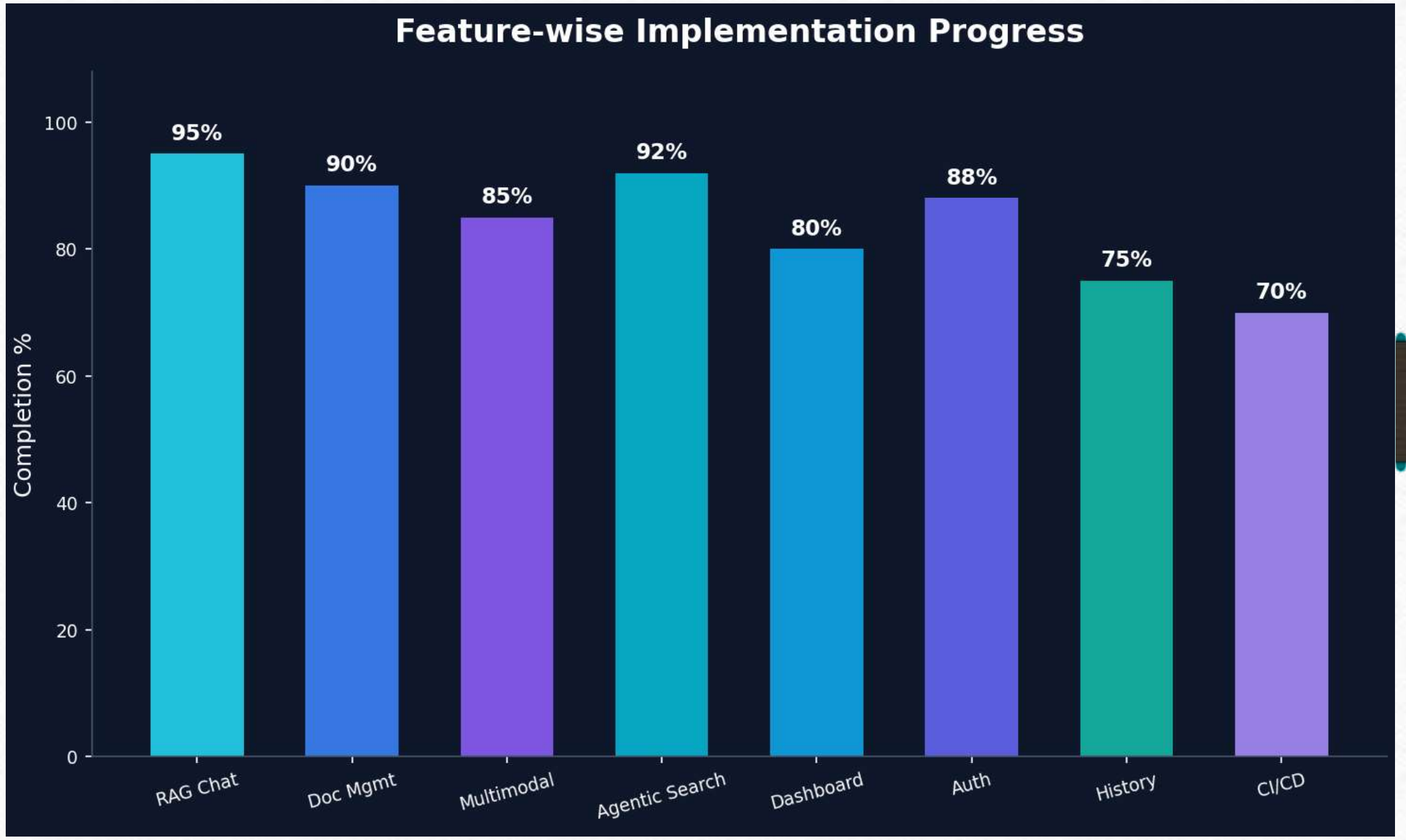
Completion: 75%

Milestones Achieved:

- ✅ Full RAG pipeline with agentic reasoning
- ✅ Multimodal document & image processing
- ✅ Complete frontend (Dashboard, Chat, Documents, History)
- ✅ JWT auth & role-based access
- ✅ Docker & CI/CD pipelines

Future Work:

- Performance optimization
- Advanced analytics
- Cloud deployment



Conclusion

Summary:

- Successfully designed and developed an enterprise-grade AI-Powered RAG Chatbot enabling natural language querying of company documents with cited, accurate responses.

- Implemented cutting-edge techniques: agentic ReAct reasoning, 3072-dimensional vector search, multimodal AI, and SSE streaming.

- Built a production-ready, containerized full-stack application with React frontend, Django backend, and intelligent RAG pipeline powered by Google Gemini.

Final Outcome:

- The system demonstrates significant improvement in document retrieval accuracy over traditional keyword search methods, making enterprise knowledge instantly accessible through natural conversation.

Implementations

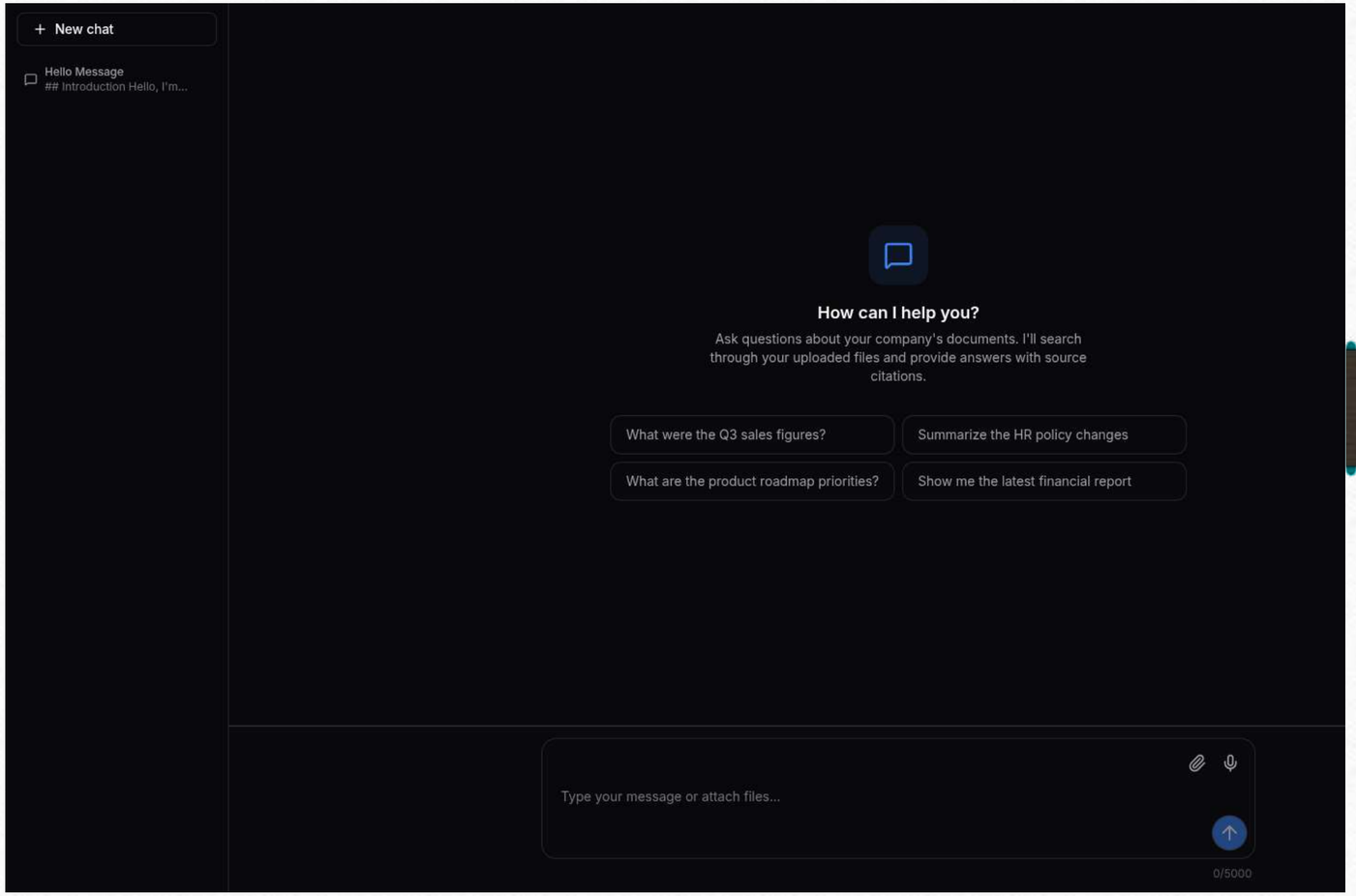
Key Implementation Highlights:

1. **RAG Pipeline (rag_chain.py)**
   - Agentic ReAct loop — max 3 iterations
   - SSE token-by-token streaming to UI
   - LLM retry on 429 with exponential backoff

2. **Vector Store (vector_store.py)**
   - Qdrant with 3072-dim Gemini embeddings
   - Hybrid semantic + metadata-scoped search

3. **Document Ingestion**
   - PDF / DOCX / CSV / Image auto-chunking
   - Gemini Vision for charts & images

4. **Real-Time Chat (SSE Streaming)**
   - Thought / Answer stream split to frontend
   - JWT auth + role-based session isolation



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Test Cases

TC-ID	Test Case Description	Test Steps (Summary)	Expected Result	Status
TC-01	Document Upload	Upload a PDF via /api/documents/	File chunked & vectors stored in Qdrant	✅ PASS
TC-02	RAG Query Response	Send query 'What is our leave policy?'	Cited answer returned with source references	✅ PASS
TC-03	Agentic Search Loop	Submit complex multi-step question	Answer produced in ≤ 3 ReAct iterations	✅ PASS
TC-04	JWT Authentication	Call API with an expired/invalid token	401 Unauthorized returned immediately	✅ PASS
TC-05	SSE Streaming	Open chat and send a message	Tokens streamed in real-time to browser	✅ PASS
TC-06	Rate Limit Retry	Simulate 429 from Gemini API	Exponential backoff + auto-retry succeeds	✅ PASS
TC-07	Multimodal Image Upload	Upload a PNG chart image	Gemini Vision extracts & indexes content	✅ PASS
TC-08	Collection Scoping	Query within a specific document collection	Results limited to that collection only	✅ PASS